# On citation quality of Language Models

Towards a framework for citation evaluation results for language models

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**Abstract**

Language Models (LMs) generate answers to queries based on their training data. However, when asked to generate responses with attribution (citation), LMs struggle to produce effective results. A significant amount of research has been conducted and is ongoing to develop effective techniques to generate attributed responses by LMs. In addition, several standardized datasets are available to assess the quality of attribution. This paper focuses on fine-tuning a small LM and compares its achieved citation quality with that of Large Language Models (LLMs) based on previous work. This paper also proposes a generic approach to tabulate results given a LM, parameters, compute resource, and a list of standardized datasets[1]. The results show that small LMs can be fine-tuned to achieve attribution quality to that of LLMs. For future work, an enhancement of the generic approach to categorize and compare citation quality results is proposed.

*KEYWORDS: attributable text generation, Language Model, citation evaluation metrics*

## 1 Introduction

LLMs generate natural language responses when prompted with a query. The use of this technology has grown exponentially due to recent advancements in generative LMs. However, a major drawback is that these models often produce responses that contain incorrect or outdated facts and are prone to hallucinations. Additional concerns include copyright issues, privacy risks, and the difficulty in assessing the credibility of the data used to produce the responses, which raises questions about the trustworthiness of LLM outputs [1].

LLM's ability to respond accurately depends heavily on the quality of its training data, and users should be able to verify the validity of the generated response. While models such as ChatGPT [2] can provide citations when prompted explicitly to do so, the generated text with citations is often inaccurate or irrelevant. Consequently, it would be valuable if LLMs could generate responses that include reliable citations alongside their content [1, 3, 4, 5].

Researchers have explored various approaches to address the challenges of attributed generation of text by LMs. Retrieval-Augmented Generation (RAG) [6, 7, 8, 9] is one such method, where the model retrieves a set of relevant external knowledge sources before generating an answer. However, RAG does not fully solve the problem, as RAG cannot guarantee that the retrieved documents are both relevant and properly used in the final response by the LM. Recent studies [6, 10, 11] propose enhancing generated answers by LMs with in-text citations that directly point to the sources of specific claims; a method known as attributable text generation, enabling users to fact check generated responses more effectively.

This work draws inspiration from previous studies and implements an Fine-Grained Reward-Based Reranker using Rejection Sampling (FGRS) algorithm that ranks multiple candidate answers generated from retrieved documents and applies fine-grained automatic evaluation functions [6, 11]. Fine-grained automatic evaluation functions provide localized (sentence-level) and specialized (citation quality) reward signals to guide model fine-tuning.

Given the problem of attributable generation for LMs, this paper tries to answer the following Research questions (RQ):

**RQ1** Can a relatively small LM[2] be fine-tuned and

[1] Code is available at https://github.com/ShabrinaSharmin/citation-quality-evaluation-task-server

[2] Small LM in this context means a LM with parameters between 5 - 10 billion.

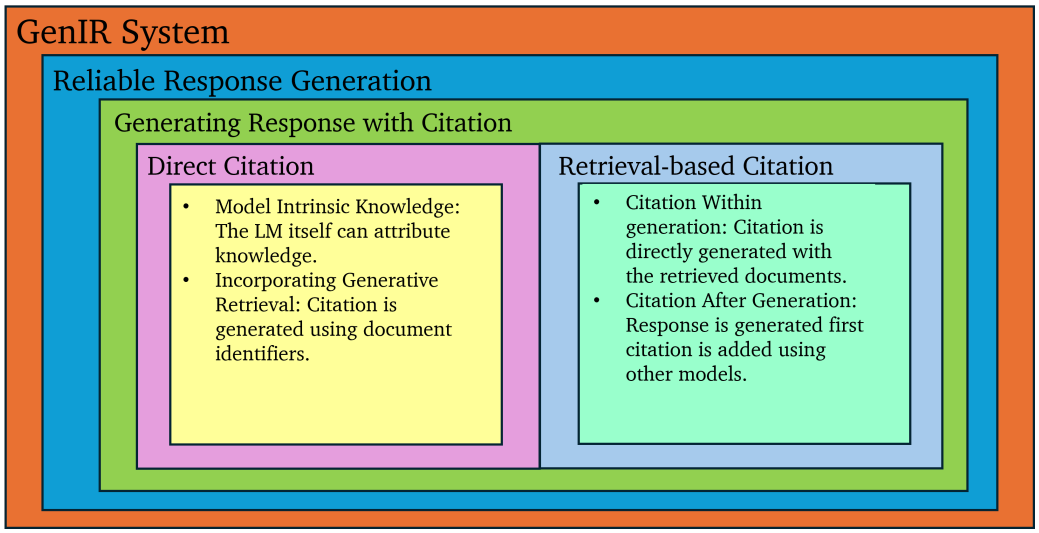



Figure 1: The figure illustrates the hierarchical taxonomy of research of "Generating Response with Citation" in the field of Generative Information Retrieval [5].

trained to attribute generated text at the same level as a LLM with more than 100 billion parameters such as ChatGPT [2] in the context of citation within generation?

**RQ2** Given the same training dataset and the same model parameters, do different models achieve different results for the quality of citation?

**RQ3** Can citation quality related evaluation results of LMs with standardized datasets be tabulated and published in a standard way?

This paper is organized as follows: In Chapter 2, background and related work are presented to introduce concepts and past work on which this paper builds upon. Chapter 3 explains the methodology followed to answer **RQ1**, **RQ2**, and **RQ3**. The results of the experiments are presented in Chapter 4, and Chapter 5 provides the analysis of the results and how it addresses **RQ1**, **RQ2**, and **RQ3**. Limitations of the experiments performed and the results presented are described in Chapter 6. And finally, the conclusion is presented in Chapter 7; related future work is proposed in Chapter 8.

# 2 Background & Related Work

## 2.1 Background

Generative Information Retrieval (GenIR) systems that incorporate the use of pre-trained LMs are capable of achieving the tasks that traditional Information Retrieval (IR) systems can perform [5]. Traditional IR systems incorporate an encoder, index pool, and similarity matching to produce a list of results ranked according to the order of relevance to the search query. In contrast, GenIR systems utilize Transformer based pre-trained LMs with natural language prompting and generative capabilities providing tailored responses for IR tasks. Thus, Generative Information Retrieval systems are gaining popularity and are giving rise to a new paradigm to achieve IR goals through generative approaches.

Li et al.'s [5] survey paper on Generative Information Retrieval categorizes the research of GenIR systems into two patterns, namely **Generative Document Retrieval** and **Reliable Response Generation (RRG)**. The focus of this paper is on the RRG pattern.

RRG is primarily concerned with strategies that help in enhancing the reliability of generated responses by LMs. However in practice, it is often the case that the responses generated by LMs are not reliable and may provide irrelevant responses. As well, the generated response may contradict the factual information or cite outdated information. One reliability enhancing aspect of RRG is to incorporate the source of information within the generated text response for a particular user query. Li et al. [5] designates the mechanism of incorporating citation as part of generated text as "Generating Response with Citation" within RRG in their taxonomy of research on GenIR.

Li et al. [5] further breaks down the field of "Generating Response with Citation" into two categories depending on how the citation mechanism was achieved. (i) "Direct Generating Response with Citation" is defined as the mechanism to utilize the LM's intrinsic memory to generate citation and thereby reliance on a retrieval module is not required. Attribution is achieved by either including source phrases in the generated text or by adding a mechanism to segment the corpus with document identifiers and using the segmented corpus for text generation. (ii) "Retrieval-based citation" is defined as the mecha-

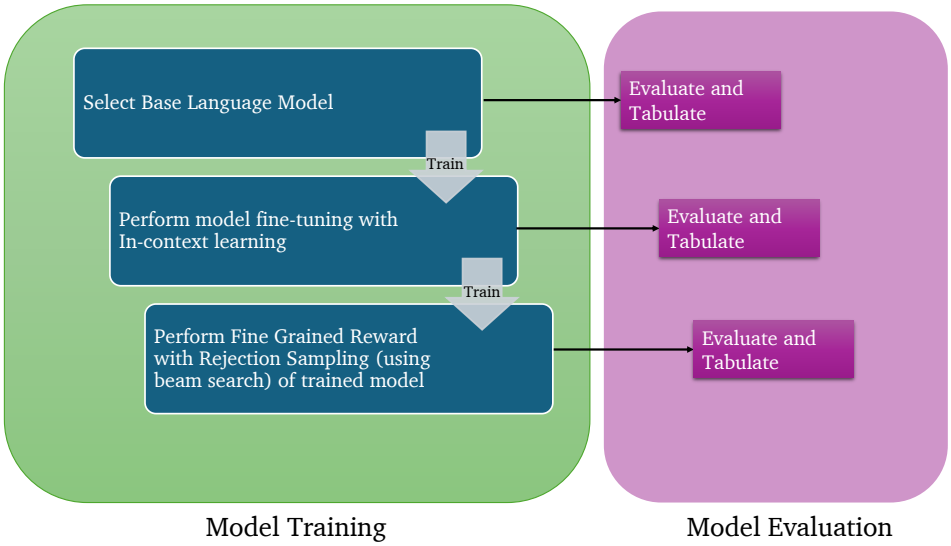



Figure 2: The figure illustrates the method established for answering **RQ1**.

nism to fetch relevant documents and use them as the source for generating responses with citation. Attribution is achieved by either incorporating citations directly within the generated text using the retrieved documents or by adding citations to the generated text after generation through other models. The taxonomy is further illustrated in Figure 1.

### 2.2 Related Work

This paper focuses on the topic of "Citation within Generation" and builds upon the work done by Gao et al. [11] and Huang et al. [6].

Gao et al. [11] propose Automatic LLM's Citation evaluation (ALCE), which includes metrics to evaluate the quality of the citation of responses generated by LMs. ALCE makes use of three QA datasets to perform the evaluation: ASQA, QAMPARI and ELI5. ALCE's citation quality evaluation measure is based on (i) correctness, which checks whether the response is accurate compared to a ground truth answer, (ii) citation recall, which determines whether cited passages entirely support the output, and (iii) citation precision, which identifies any irrelevant citations.

Soft prompt-based fine-tuning (SFT) [12] within additive fine-tuning is a parameter-efficient fine-tuning method for pretrained LMs where soft prompts or prefix vectors are appended to input embeddings or hidden states of LMs during fine-tuning [13]. Huang et al. [6] uses SFT with in-context learning from LLM responses to fine-tune small LMs, which is referred to as a distilled model or $\mathcal{M}_{dist}$ and the process is termed **model distillation**. $\mathcal{M}_{dist}$ is used with FGRS [14, 10, 5] to further train the model. Huang et al. [6] can demonstrate that SFT LMs ($\mathcal{M}_{dist}$) ranks better in terms of citation quality than the base LM, and $\mathcal{M}_{dist}$ when trained with FGRS performs better than $\mathcal{M}_{dist}$.

## 3 Methodology

### 3.1 Problem Definition

The problem of attributable text generation remains similar to [11]. Formally, given a query $q$ and a corpus $\mathcal{D}$, a LM generates answers $\mathcal{A}$ which consists of $\mathcal{N}$ sentences where $\mathcal{N} = \{s_1, s_2, \cdots, s_n\}$ along with a set of references $C = \{[1], [2], \cdots [n]\}$, where each reference $C_i \subseteq \mathcal{D}$. The correctness of the attribution can be quantified by checking that the relationship $C_i \subseteq \mathcal{D}$ holds.

### 3.2 Evaluation metrics

1. **Correctness** This metric measures whether the generated responses are accurate and consistent with the gold-standard answers. The implementation of this metric varies across datasets. In this study, we use EM Rec, Rec-5, and Claim Rec (as described in [11]) to quantify correctness. The metric captures the proportion of key information pieces from the gold standard that are included in the generated responses.

2. **Citation precision** Citation precision evaluates whether the sources cited in the generated responses are correct and relevant. Precision measures whether the citations themselves are relevant and necessary. High citation precision reduces the

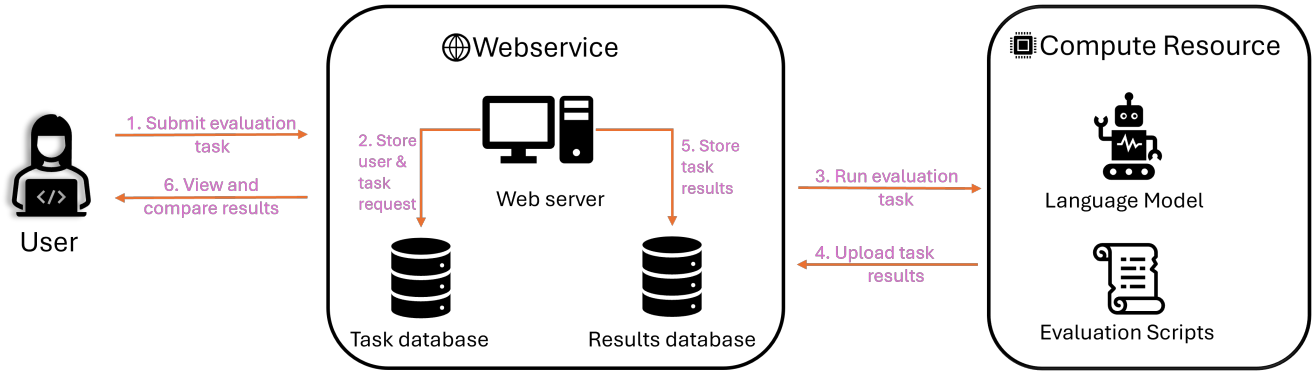



Figure 3: The figure illustrates the system proposed for **RQ3**.

need for users to manually verify sources and is critical for user trust and satisfaction.

3. **Citation recall** Citation recall measures the proportion of relevant sources that are actually included as citations in the generated response. Recall measures whether the sentences in the generated text properly support the cited passages or not. This metric ensures that the answers generated by the LMs are supported by relevant references, contributing to both the quality and reliability of the responses.

### 3.3 Datasets

ALCE [11] citation evaluation uses the following datasets:

1. **ASQA** Long-form questions and answers with ambiguous questions requiring synthesis across multiple passages.

2. **QAMPARI** Factoid questions and answers where answers are lists of entities and require multi-passage integration.

3. **ELI5** Long-form explanatory question and answers ("Why/How/What" questions) with natural language responses.

Each dataset has its own passage corpus, and the top-5 relevant passages are retrieved using Generalizable T5-based Dense Retrievers (GTR) or Best Matching 25 (BM25), ensuring responses are grounded in external sources for correctness and citation evaluation.

### 3.4 Method for RQ1

Huang et al. [6] provide evidence supporting the claim that small LMs can be effectively trained using SFT and FGRS to perform at the same level as large LLMs. However, the results show the claim to be true using only *one* small LM. Numerous open-source LMs are available on which SFT and FGRS can be performed, and the quality of citation evaluated. The method is presented in Figure 2.

### 3.5 Method for RQ2

LLaMA2 7B [14] is used as the small LM by Huang et al. [6]. Mistral 7B [15] with the same number of model parameters is selected for experimentation according to the method described in Section 3.4. It should be noted that numerous metrics for Mistral 7B have been published, claiming better performance than LLaMA2 7B [15]. However, no citation metrics are reported.

### 3.6 Method for RQ3

There is an effort in the literature to establish metrics for citation evaluation [11, 3]. However, the reporting of results remains fragmented across studies. Different research groups often evaluate on different datasets (ASQA, QAMPARI, ELI5, ExpertQA, etc.), using varied setups and reporting formats [11, 6], which makes a direct comparison between models difficult. Currently, there is no standardized service or framework that can tabulate and publish results in a consistent way. Establishing such a framework would allow researchers to compare models fairly, identify performance gaps, and accelerate progress in developing reliable citation-capable language models. A web service is proposed that would cover the use-cases:

**UC1** Users can select a base model, an evaluation type (ALCE, ExpertQA, etc.), specify computing resources, and submit an evaluation pipeline. Users are notified by email once the evaluation pipeline task is completed.

**UC2** Users can view the results of their past and current evaluation pipeline, and compare results using multiple view layouts.

Figure 3 details the web service.

| | ASQA | | | QAMPARI | | | | ELI5 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Correctness | Citation | | Correctness | | Citation | | Correctness | Citation | |
| | EM Rec | Rec | Prec | Rec.-5 | Prec | Rec | Prec | Claim Rec | Rec | Prec |
| ICL ChatGPT [1] | 39.96 | 74.72 | 70.97 | 18.34 | 18.57 | 17.65 | 20.60 | 13.47 | 50.94 | 47.58 |
| $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ | 32.65 | 72.13 | 69.25 | 12.34 | 13.16 | 12.85 | 16.76 | 9.83 | 49.75 | 48.33 |
| +FGRS | 34.93 | 77.07 | 71.49 | 16.8 | 18.08 | 16.01 | 17.44 | - | - | - |
| $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ | 32.90 | 73.89 | 68.60 | 15.98 | 13.97 | 13.68 | 15.34 | 12.6 | 49.91 | 51.09 |
| +FGRS | 34.77 | 76.44 | 74.05 | 16.02 | 16.20 | 15.16 | 16.626 | - | - | - |

[1] Baseline from [6]

Table 1: Performance comparison across models for ASQA, QAMPARI, and ELI5 datasets in ALCE[11]. Units are in percentages.

# 4 Results

Table 1 reports the results of the experiments for **RQ1**, and **RQ2**. Table 1 includes LLM ChatGPT[2] with in-context learning (ICL) as baseline taken from [6] highlighted in blue.

Figures 4, 5, and 6 show the Citation Quality Evaluation Task Server (CQETS) web service for **RQ3**. Figure 4a shows the login page. The dashboard is visualized in Figure 4b. A page to view user tasks is shown in Figure 5a. Adding a new pipeline task is illustrated in 5b. Figure 6 presents the page where the citation evaluation metrics of the completed tasks can be compared within the web service.

# 5 Analysis

The first row of Table 1 presents the results for a LLM (ICL ChatGPT) taken from [6] and is used as the baseline to analyze **RQ1**. Evaluation of the ELI5 dataset for both $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS and $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS cannot be reported due to computing resource limitations as described in Section 6.1.

## 5.1 Analysis of RQ1

The results in Table 1 indicate that relatively small LMs (7B parameters) can achieve text attribution quality comparable to, and in certain cases exceeding, that of LLMs (>100B parameters).

On ALCE ASQA dataset, $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ achieved a citation precision value of 69.25, which increased to **71.49** after applying FGRS. $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ demonstrated a similar trend, improving from 68.90 to **74.05** precision following FGRS. $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS and $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS report better citation precision than ChatGPT.

On ALCE QAMPARI dataset, $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS achieved a citation precision of 17.44 and $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS reported a citation precision of 16.63 compared to ChatGPT's 18.57.

On ALCE ELI5 dataset, both $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ and $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ outperformed ChatGPT's citation precision of 47.58. $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ reported 48.33 and $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ reported 51.09. Although both $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ and $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ with FGRS achieve close to or higher citation precision compared to ChatGPT, the correctness measure is lower for both soft fine-tuned small LM.

It can be demonstrated from the results in Table 1 that small LMs with SFT can achieve text attribution close to or even better than LLM. Table 2 highlighting the differences is presented in Appendix A3.

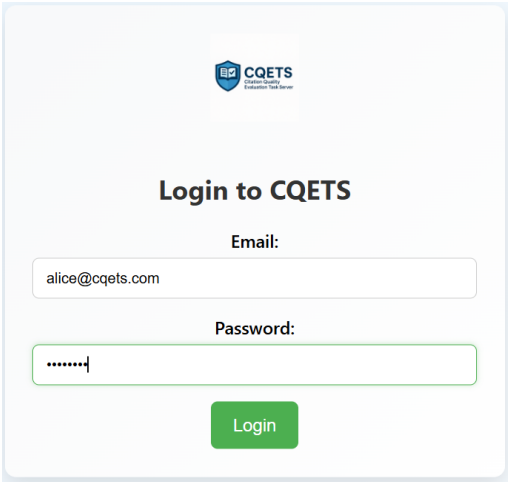
## 5.2 Analysis for RQ2

The following comparison of the results from Table 1 are paired in the analysis below: $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ and $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$; $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS and $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS.

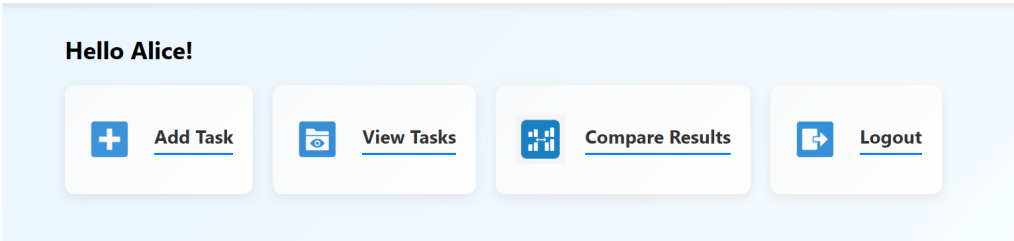
On ALCE ASQA dataset, $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ reports a slightly higher citation precision, 69.25, compared to 68.60 achieved by $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$. But $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS surpasses $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS with 2.5 percentage points.

On ALCE QAMPARI dataset, $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ outperforms $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ by 1.42 percentage points, while the delta between $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS and $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS is very small or insignificant.

On ALCE ELI5 dataset, $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ achieves better citation precision compared to $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ by 2.77 percentage points.




(a) CQETS Login Page




(b) CQETS Dashboard

Figure 4: CQETS Login and Dashboard pages

From a citation recall perspective, $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ ranks better compared to $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$. Whereas, $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS reports slightly better results than $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS.

These results show that different small LMs achieve different results for the same dataset and suggest other factors such as LMs architecture contributing to the results. $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ ranks better in the citation precision-related metrics while $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ ranks better in the citation recall-related metrics. Tables 3 and 4 that highlight the differences are presented in Appendix A3.

## 5.3 Analysis for RQ3

A web service CQETS is designed as illustrated in Figures 4, 5, and 6 covering the use-cases described in Section 3.6. The web service can be used as a tool for collecting citation evaluation results. Some limitations of the web service are described in Section 6.2 Initially, a login page is displayed. Once the credentials are validated, the user is greeted with a dashboard as shown in Figure 4b with the following options: (i) Add Task, (ii) View Tasks, (iii) Compare Results, and (iv) Logout as shown in Figure 4b.

### 5.3.1 CQETS: Add Task

The web page to add a task is shown in Figure 5b. The user fills in the following details about the task: the name of the task, the model to be used from Hugging Face [16] on which the evaluation will be run, the necessary keys for accessing Hugging face and Wandb.ai [17], and selecting the evaluation type. The page also contains fields to enter information about the computing resources to be used such as an IP address, username, and key. Lastly, the user can enter a additional set of email addresses that will be used to send notifications about the progress of the task once submitted. Once the task is submitted, the user is presented with a task summary page, and the user can navigate back to any of the web pages using the navigation bar.

### 5.3.2 CQETS: View Task

The web page for viewing tasks is shown in Figure 5a. The web page lists all tasks submitted by the user in a tabular format, including the status of the tasks. If a task has finished successfully and the results are available, the user is presented with an option to see the collected results from the compute resource.

## All Tasks

| TASK NAME | MODEL | EVALUATION | COMPUTE IP | USER | ADDITIONAL EMAILS | STATUS | RESULTS |
|---|---|---|---|---|---|---|---|
| Task 1 | mistralai/Mistral-7B-v0.3 | ALCE | 192.168.0.1 | alice | alice.team@example.com | Complete | View Results |
| Task 2 | meta-llama/Llama-2-7b-hf | ALCE | 192.168.0.2 | alice | qa@example.com | Complete | View Results |
| Task 3 | mistralai/Mistral-7B-v0.3 (Finetuned_fgRS) | ALCE | 192.168.0.3 | alice | | Complete | View Results |
| Task 45 | meta-llama/Llama-2-7b-hf (Finetuned_fgRS) | ALCE | 192.168.0.3 | alice | | Complete | View Results |
| Task 45 | meta-llama/Llama-2-7b-hf | ALCE | 192.168.0.3 | alice | | Not Started | - |

(a) CQETS View Task page

CQETS

## Add New Task

Task Details

**Task Name:**

Enter task name

**Model Name on Hugging Face:**

Enter model name

Open HF

**Hugging Face Key:**

Enter HF key

**Wandb.ai Key:**

Enter Wandb.ai key

**Evaluation Type:**

ALCE

Specify Compute Resource

**Compute Resource IP Address:**

Enter IP address

**User Login Name:**

Enter login name

**Compute Resource Certificate:**

Paste certificate text here

Task Notification Results

**Additional Email Addresses (separate with ";" ):**

email1@example.com;email2@example.com

Submit Task

(b) CQETS Add Task page

Figure 5: CQETS View Task and Add Task pages

## Compare Results

**Select Result 1**

1 | mistralai/Mistral-7B-v0.3

**Select Result 2**

2 | meta-llama/Llama-2-7b-hf

## Comparison Table

**Note: Deltas are shown with respect to Result 1.**
**if Result 1 sub-metric is smaller than Result 2, the delta is highlighted as RED**
**if Result 1 sub-metric is greater than Result 2, the delta is highlighted as GREEN**

### ASQA

| SUB-METRIC | RESULT 1 MISTRALAI/MISTRAL-7B-V0.3 | RESULT 2 META-LLAMA/LLAMA-2-7B-HF | DELTA |
|---|---|---|---|
| Correctness EM_REC | 32.90436005625879 | 32.64767932489451 | 0.25668073136427694 |
| Citation Rec | 73.89767932489451 | 72.12653204741812 | 1.7711472774763877 |
| Citation Prec | 68.60659031545107 | 69.25406871609404 | 0.6474784006429672 |

### ELI5

| SUB-METRIC | RESULT 1 MISTRALAI/MISTRAL-7B-V0.3 | RESULT 2 META-LLAMA/LLAMA-2-7B-HF | DELTA |
|---|---|---|---|
| Correctness Claims Rec | 12.6 | 9.833333333333332 | 2.7666666666666675 |
| Citation Prec | 49.911278998779004 | 49.7525 | 0.15877899877900603 |
| Citation Rec | 51.086137566137566 | 48.333492063492066 | 2.7526455026454997 |

### QAMPARI

| SUB-METRIC | RESULT 1 MISTRALAI/MISTRAL-7B-V0.3 | RESULT 2 META-LLAMA/LLAMA-2-7B-HF | DELTA |
|---|---|---|---|
| Correctness Req.-5 | 15.979999999999997 | 12.340000000000002 | 3.6399999999999952 |
| Correctness Prec | 13.966975024508042 | 13.1616113489194 | 0.8053636755886409 |
| Citation Rec | 13.68258975286839 | 12.851412204179063 | 0.8311775486893271 |
| Citation Prec | 15.340996679275317 | 16.763056649233118 | 1.4220599699578003 |

Figure 6: CQETS Compare Results page

#### 5.3.3 CQETS: Compare Results

The Web page to compare the results of the completed tasks is shown in Figure 6. The page allows the user to pick two completed tasks and compare the results side by side listing the delta between them. Positive deltas are colored green, and negative deltas are colored red.

#### 5.3.4 CQETS: Logout

The logout option is presented to the user on the dashboard page and on the navigation bar. Once the user clicks on the logout link, the log-in page is displayed.

# 6 Limitations

## 6.1 Computing resources

The computing environment used for the experiments is listed in A1. Amazon EC2 cloud instance with 48 GB graphics processing capacity was used, which was not sufficient for fine-tuning of the LMs with certain training datasets. The results in Table 1 mark it with a "-" for inconclusive experiments due to insufficient computing resources.

## 6.2 Web service limitations

The web service developed has the following limitations:

1. Only a single type of citation evaluation ALCE is supported.
2. Currently, there is no mechanism to connect the web service with compute resources.
3. The data for the tasks and the results are stored in a JSON format [18].

# 7 Conclusion

The paper investigated the capacity of small LMs to perform attributable text generation, with focus on in-text citation quality. Three research questions guided the study: (i) whether small LMs can be trained to achieve the same level of text attribution as that of LLMs, (ii) whether models with the same number of parameters achieve different levels of citation quality when trained under identical conditions, and (iii) whether evaluation results related to citation quality can be tabulated and published in a standard way.

The findings in this paper demonstrate that small LMs, when fine-tuned with FGRS, can indeed reach or surpass the quality achieved by much larger LMs. Mistral 7B and LLaMA2 7B models, when distilled, exceeded the citation precision of ChatGPT, despite having far fewer parameters. These results provide strong evidence to support **RQ1** which confirms that properly trained small LMs can close the gap with proprietary large-scale LLM in terms of citation reliability.

With respect to **RQ2**, the analysis revealed consistent differences between models under the same training setup. LLaMA2 7B exhibited higher citation recall scores across datasets, while Mistral 7B achieved higher citation precision scores. These results suggest that model architecture and pretraining corpus influence how effectively LM performs on citation recall and precision. Thus, RQ2 is affirmed: different models display distinct trade-offs in attribution quality even when fine-tuned under the same conditions.

On **RQ3**, a web service is developed to demonstrate a framework that can be used to tabulate and compare results of citation quality of different LMs using standardized datasets.

# 8 Future Work

The citation quality related evaluation performed in this study using standardized datasets can be extended for similar small LM and tabulated further to validate generalizability. In addition, further studies can be done to understand whether certain LM architecture performs better text attribution while keeping model parameters constant. The proposed web service can be further adapted to include other benchmark evaluation types. Comparison of results from evaluations can be further enhanced with visual aids such as metrics comparison charts.

## Acronyms

**ALCE** Automatic LLM's Citation evaluation. 3–5, 9

**CQETS** Citation Quality Evaluation Task Server. 5–9

**FGRS** Fine-Grained Reward-Based Reranker using Rejection Sampling. 1, 3–6, 9, 13

**GenIR** Generative Information Retrieval. 2

**IR** Information Retrieval. 2

**LLM** Large Language Model. 1–5, 9

**LM** Language Model. 1–6, 9, 13

**RAG** Retrieval-Augmented Generation. 1

**RQ** Research question. 1

**RRG** Reliable Response Generation. 2

**SFT** Soft prompt-based fine-tuning. 3–5

## References


[1] M. Khalifa, D. Wadden, E. Strubell, H. Lee, L. Wang, I. Beltagy, and H. Peng, "Source-aware training enables knowledge attribution in language models," 2024.

[2] OpenAI, "Chatgpt." *Large language model.*, 2023.

[3] H. Rashkin, V. Nikolaev, M. Lamm, L. Aroyo, M. Collins, D. Das, S. Petrov, G. S. Tomar, I. Turc, and D. Reitter, "Measuring attribution in natural language generation models," 2022.

[4] J. Huang and K. C.-C. Chang, "Citation: A key to building responsible and accountable large language models," 2024.

[5] X. Li, J. Jin, Y. Zhou, Y. Zhang, P. Zhang, Y. Zhu, and Z. Dou, "From matching to generation: A survey on generative information retrieval," 2025.

[6] C. Huang, Z. Wu, Y. Hu, and W. Wang, "Training language models to generate text with citations via fine-grained rewards," 2024.

[7] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang, "Realm: Retrieval-augmented language model pre-training," 2020.

[8] A. Piktus, F. Petroni, V. Karpukhin, D. Okhonko, S. Broscheit, G. Izacard, P. Lewis, B. Oğuz, E. Grave, W. tau Yih, and S. Riedel, "The web is your oyster - knowledge-intensive nlp against a very large web corpus," 2022.

[9] G. Izacard, P. Lewis, M. Lomeli, L. Hosseini, F. Petroni, T. Schick, J. Dwivedi-Yu, A. Joulin, S. Riedel, and E. Grave, "Atlas: Few-shot learning with retrieval augmented language models," 2022.

[10] R. Nakano, J. Hilton, S. Balaji, J. Wu, L. Ouyang, C. Kim, C. Hesse, S. Jain, V. Kosaraju, W. Saunders, X. Jiang, K. Cobbe, T. Eloundou, G. Krueger, K. Button, M. Knight, B. Chess, and J. Schulman, "Webgpt: Browser-assisted question-answering with human feedback," 2022.

[11] T. Gao, H. Yen, J. Yu, and D. Chen, "Enabling large language models to generate text with citations," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (H. Bouamor, J. Pino, and K. Bali, eds.), (Singapore), pp. 6465–6488, Association for Computational Linguistics, Dec. 2023.

[12] X. L. Li and P. Liang, "Prefix-tuning: Optimizing continuous prompts for generation," 2021.

[13] L. Xu, H. Xie, S.-Z. J. Qin, X. Tao, and F. L. Wang, "Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment," 2023.

[14] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale, D. Bikel, L. Blecher, C. C. Ferrer, M. Chen, G. Cucurull, D. Esiobu, J. Fernandes, J. Fu, W. Fu, B. Fuller, C. Gao, V. Goswami, N. Goyal, A. Hartshorn, S. Hosseini, R. Hou, H. Inan, M. Kardas, V. Kerkez, M. Khabsa, I. Kloumann, A. Korenev, P. S. Koura, M.-A. Lachaux, T. Lavril, J. Lee, D. Liskovich, Y. Lu, Y. Mao, X. Martinet, T. Mihaylov, P. Mishra, I. Molybog, Y. Nie, A. Poulton, J. Reizenstein, R. Rungta, K. Saladi, A. Schelten, R. Silva, E. M. Smith, R. Subramanian, X. E. Tan, B. Tang, R. Taylor, A. Williams, J. X. Kuan, P. Xu, Z. Yan, I. Zarov, Y. Zhang, A. Fan, M. Kambadur, S. Narang, A. Rodriguez, R. Stojnic, S. Edunov, and T. Scialom, "Llama 2: Open foundation and fine-tuned chat models," 2023.

[15] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. L. Scao, T. Lavril, T. Wang, T. Lacroix, and W. E. Sayed, "Mistral 7b," 2023.

[16] Hugging Face, "Hugging face – the ai community building the future.." https://huggingface.co/. Accessed: 2025-08-24.

[17] Weights & Biases, "Weights & biases." https://wandb.ai/. Accessed: 2025-08-24.

[18] T. Bray, "The javascript object notation (json) data interchange format." RFC 8259, 2017.

[19] Amazon Web Services, Inc., "Amazon elastic compute cloud (ec2)." https://aws.amazon.com/ec2/, 2025. Accessed: 2025-08-20.

[20] Amazon Web Services, Inc., "Amazon ec2 g6 instances." https://aws.amazon.com/ec2/instance-types/g6/, 2025. Accessed: 2025-08-20.

# Appendix

## A1 Computing environment

Amazon EC2 [19] provides scalable virtual cloud instances with multiple options for Graphics Processing Units with different pricing per hour. Amazon EC2 g6e.2xlarge[20] was used to perform the experiments for the results tabulated in Table 1.
Details of the computing resource used:

```
---------------------------------------------------
ubuntu@ip-XXX-XX-XX-XX:~$ neofetch
ubuntu@ip-XXX-XX-XX-XX
OS: Ubuntu 22.04.5 LTS x86_64
Host: g6e.2xlarge
Kernel: 6.8.0-1033-aws
Uptime: 3 mins
Packages: 1298 (dpkg), 6 (snap)
Shell: bash 5.1.16
Resolution: 800x600
Terminal: /dev/pts/0
CPU: AMD EPYC 7R13 (8) @ 2.949GHz
GPU: NVIDIA 30:00.0 NVIDIA Corporation Device 26b9
Memory: 636MiB / 63426MiB
---------------------------------------------------
```

Details on the graphics processing unit resources:

```
ubuntu@ip-XXX-XX-XX-XX:~$ nvidia-smi
Wed Aug 27 00:16:17 2025
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA L40S                    On  |   00000000:30:00.0 Off |                    0 |
| N/A   29C    P8             24W /  350W |       0MiB /  46068MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

## A2 Evaluation script

The evaluation scripts from https://github.com/HCY123902/atg-w-fg-rw [6] were modified to run distillation, inference, and evaluation on Mistral 7B.

## A3 Results Comparison

| | ASQA | | | QAMPARI | | | | ELI5 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Correctness | Citation | | Correctness | | Citation | | Correctness | Citation | |
| | EM Rec | Rec | Prec | Rec.-5 | Prec | Rec | Prec | Claim Rec | Rec | Prec |
| ICL ChatGPT [1] | 39.96 | 74.72 | 70.97 | 18.34 | 18.57 | 17.65 | 20.60 | 13.47 | 50.94 | 47.58 |
| $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ | 32.65 | 72.13 | 69.25 | 12.34 | 13.16 | 12.85 | 16.76 | 9.83 | 49.75 | 48.33 |
| +FGRS | 34.93 | 77.07 | 71.49 | 16.8 | 18.08 | 16.01 | 17.44 | - | - | - |
| $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ | 32.90 | 73.89 | 68.60 | 15.98 | 13.97 | 13.68 | 15.34 | 12.6 | 49.91 | 51.09 |
| +FGRS | 34.77 | 76.44 | 74.05 | 16.02 | 16.20 | 15.16 | 16.626 | - | - | - |

Table 2: Performance comparison of baseline model (ChatGPT) with distilled small LMs and distilled+FGRS small LMs

| | ASQA | | | QAMPARI | | | | ELI5 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Correctness | Citation | | Correctness | | Citation | | Correctness | Citation | |
| | EM Rec | Rec | Prec | Rec.-5 | Prec | Rec | Prec | Claim Rec | Rec | Prec |
| $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ + FGRS | 34.93 | 77.07 | 71.49 | 16.8 | 18.08 | 16.01 | 17.44 | - | - | - |
| $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ + FGRS | 34.77 | 76.44 | 74.05 | 16.02 | 16.20 | 15.16 | 16.626 | - | - | - |

Table 3: Performance comparison between distilled small LMs

| | ASQA | | | QAMPARI | | | | ELI5 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Correctness | Citation | | Correctness | | Citation | | Correctness | Citation | |
| | EM Rec | Rec | Prec | Rec.-5 | Prec | Rec | Prec | Claim Rec | Rec | Prec |
| $\mathcal{M}_{\text{dist(LLaMA-2-7B)}}$ | 32.65 | 72.13 | 69.25 | 12.34 | 13.16 | 12.85 | 16.76 | 9.83 | 49.75 | 48.33 |
| $\mathcal{M}_{\text{dist(Mistral-7B-v0.3)}}$ | 32.90 | 73.89 | 68.60 | 15.98 | 13.97 | 13.68 | 15.34 | 12.6 | 49.91 | 51.09 |

Table 4: Performance comparison between small LMs that are distilled and trained with FGRS